



Darshan UNIVERSITY

Python Programming - 2301CS404

Lab - 5

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1) Character Category Counter

Problem: Count uppercase, lowercase, digits, and special characters in a string.

Sample Inputs & Outputs:

- Input: Python3@Lab → Uppercase: 2, Lowercase: 7, Digits: 1, Special: 1
- Input: HELLO123 → Uppercase: 5, Lowercase: 0, Digits: 3, Special: 0
- Input: hi@#9 → Uppercase: 0, Lowercase: 2, Digits: 1, Special: 2

```
In [68]: P1str = "Python3@Lab"
P1str2 = "HELLO123"
P1str3 = "hi@#9"
lowerCounter = 0
upperCounter = 0
digitCounter = 0
SpecialCounter = 0

for x in P1str:
    if x.islower():
        lowerCounter+=1
    elif x.isupper():
        upperCounter+=1
    elif x.isdigit():
        digitCounter+=1
    else:
        SpecialCounter+=1

print(f"{P1str} -> Uppercase : {upperCounter} lowercase : {lowerCounter} digits

lowerCounter = 0
upperCounter = 0
```

```

digitCounter = 0
SpecialCounter = 0

for x in P1str2:
    if x.islower():
        lowerCounter+=1
    elif x.isupper():
        upperCounter+=1
    elif x.isdigit():
        digitCounter+=1
    else:
        SpecialCounter+=1

print(f"{P1str2} -> Uppercase : {upperCounter} lowercase : {lowerCounter} digits : {digitCounter} Special : {SpecialCounter}")

lowerCounter = 0
upperCounter = 0
digitCounter = 0
SpecialCounter = 0

for x in P1str3:
    if x.islower():
        lowerCounter+=1
    elif x.isupper():
        upperCounter+=1
    elif x.isdigit():
        digitCounter+=1
    else:
        SpecialCounter+=1

print(f"{P1str3} -> Uppercase : {upperCounter} lowercase : {lowerCounter} digits : {digitCounter} Special : {SpecialCounter}")

```

Python3@Lab -> Uppercase : 2 lowercase : 7 digits : 2 Special : 0

HELLO123 -> Uppercase : 5 lowercase : 0 digits : 3 Special : 0

hi@#9 -> Uppercase : 0 lowercase : 2 digits : 3 Special : 0

2) Reverse Each Word

- Input: Python is Easy → nohtyP si ysaE
- Input: Hello World → olleH dlrow
- Input: Learn Python Fast → nrael nohtyP tsaF

```

In [56]: P2str = "Python is Easy"
newStr = ' '.join(word[::-1] for word in P2str.split())
print(newStr)

P2str2 = "Hello World"
newStr2 = ' '.join(word[::-1] for word in P2str2.split())
print(newStr2)

P3str3 = "Learn Python Fast"
newStr3 = ' '.join(word[::-1] for word in P3str3.split())
print(newStr3)

```

```

nohtyP si ysaE
olleH dlrow
nrael nohtyP tsaF

```

3) Case Pattern Identifier

- PYTHON → Uppercase
- python → Lowercase
- Python Programming → Title Case

```
In [27]: P3str = "python"
P3str2 = "PYTHON"
P3str3 = "Python Programming"
print(f"lowercase: {P3str.islower()}")
print(f"uppercase : {P3str2.isupper()}")
print(f"titleCase : {P3str3.istitle()}")
```

```
lowercase: True
uppercase : True
titleCase : True
```

4) First and Last Occurrence Finder

- programming , g → First:3 Last:10
- banana , a → First:1 Last:5
- hello , z → Not found

```
In [32]: P4str = "programming"
P4str2 = "banana"
P4str3 = "hello"

print(f"{P4str} , g FirstIndex: {P4str.find('g')} , LastIndex:{P4str.rfind('g')}")
print(f"{P4str2} , a FirstIndex: {P4str2.find('a')} , LastIndex:{P4str2.rfind('a')}")
print(f"{P4str3} , z FirstIndex: {P4str3.find('z')} , LastIndex:{P4str3.rfind('z')}")
```

```
programming , g FirstIndex: 3 , LastIndex:10
banana , a FirstIndex: 1 , LastIndex:5
hello , z FirstIndex: -1 , LastIndex:-1
```

5) Word Frequency Counter (ignore the case)

- Python is easy and Python is powerful → python: 2
- Java is popular but java is verbose → java: 2
- C is fast → c:1

```
In [51]: P5str = "Python is easy and Python is powerful"
P5str = P5str.lower()
print(f"{P5str}: python {(P5str.count('python'))}")

P5str2 = "Java is popular but java is verbose"
P5str2 = P5str2.lower()
print(f"{P5str2}: java {(P5str2.count('java'))}")

P5str3 = "C is fast"
P5str3 = P5str3.lower()
print(f"{P5str3}: c {(P5str3.count('c'))}")
```

python is easy and python is powerful: python 2
 java is popular but java is verbose: java 2
 c is fast: c 1

6) Remove Extra Spaces

- Python is fun → Python is fun
- Hello World → Hello World
- NoExtraSpace → NoExtraSpace

```
In [64]: P6str = " Python is fun"
P6str2 = " Hello World"
P6str3 = "NoExtraSpace"
newStr4 = ' '.join(word[::1] for word in P6str.split())
newStr5 = ' '.join(word[::1] for word in P6str2.split())
newStr6 = ' '.join(word[::1] for word in P6str3.split())

print(f"{newStr4}")
print(f"{newStr5}")
print(f"{newStr6}")
```

Python is fun
 Hello World
 NoExtraSpace

7) Prefix and Suffix Removal

- unhappy.txt → happy
- pretest.py → test
- unwanted.doc → wanted

```
In [76]: P7str = "unhappy.txt"
P7str2 = "pretest.py"
P7str3 = "unwanted.doc"
print(f"{P7str} -> {P7str.removeprefix('un').removesuffix('.txt')}")
print(f"{P7str2} -> {P7str2.removeprefix('pre').removesuffix('.py')}")
print(f"{P7str3} -> {P7str3.removeprefix('un').removesuffix('.doc')}")
```

unhappy.txt -> happy
 pretest.py -> test
 unwanted.doc -> wanted

8) Replace Vowels

- Programming → Pr*gr*mm*ng
- Education → *d*c*t**n
- Sky → Sky

```
In [87]: P8str = "Programming"
P8str2 = "Education"
P8str3 = "Sky"
for x in P8str:
    if (x == 'a'):
        P8str = P8str.replace(x, '*')
    elif (x == 'e'):
```

```

        P8str = P8str.replace(x , '*')
    elif (x == 'i'):
        P8str = P8str.replace(x , '*')
    elif (x == 'o'):
        P8str = P8str.replace(x , '*')
    elif (x == 'u'):
        P8str = P8str.replace(x , '*')
print(f"{P8str}")

for x in P8str2:
    if (x == 'a'):
        P8str2 = P8str2.replace(x , '*')
    elif (x == 'e'):
        P8str2 = P8str2.replace(x , '*')
    elif (x == 'i'):
        P8str2 = P8str2.replace(x , '*')
    elif (x == 'o'):
        P8str2 = P8str2.replace(x , '*')
    elif (x == 'u'):
        P8str2 = P8str2.replace(x , '*')
print(f"{P8str2}")

for x in P8str3:
    if (x == 'a'):
        P8str3 = P8str3.replace(x , '*')
    elif (x == 'e'):
        P8str3 = P8str3.replace(x , '*')
    elif (x == 'i'):
        P8str3 = P8str3.replace(x , '*')
    elif (x == 'o'):
        P8str3 = P8str3.replace(x , '*')
    elif (x == 'u'):
        P8str3 = P8str3.replace(x , '*')
print(f"{P8str3}")

```

Pr*gr*mm*ng

Ed*c*t**n

Sky

9) String Compression

- aaabbccccc → a3b2c4
- xxxyyyz → x3y3z1
- abcd → a1b1c1d1

In [6]: P9str = "aaabbccc"

```

result = ""
count = 1

```

```

for i in range(len(P9str)):
    if i + 1 < len(P9str) and P9str[i] == P9str[i + 1]:
        count += 1
    else:
        result += P9str[i] + str(count)
        count = 1

```

```
print(P9str, "-> " , result)
```

aaabbccc -> a3b2c3

10) Toggle Case (without using str.swapcase())

- PyThOn → pYtHoN
- HEllO → heLLo
- Python → pYTHON

```
In [94]: P10str = "PyThOn"
P10str2 = "HEllO"
P10str3 = "Python"

print(f"{P10str} -> {P10str.swapcase()}")
print(f"{P10str2} -> {P10str2.swapcase()}")
print(f"{P10str3} -> {P10str3.swapcase()}")
```

PyThOn -> pYtHoN

HEllO -> heLLo

Python -> pYTHON

11) Username Validation

valid username rules:

1. only alphabets and digits
2. does not start with a digit
3. length ≥ 6

- User123 → Valid
- 123User → Invalid
- Us@12 → Invalid

```
In [8]: P11str = "User123"
if (P11str.isalnum()):
    print(f"{P11str}:- Valid")
else:
    print(f"{P11str}:- Invalid")

P11str2 = "123User"
if (P11str2.isdigit()):
    print(f"{P11str2}:- Valid")
else:
    print(f"{P11str2}:- Invalid")

P11str3 = "Us@12"
if (P11str3.isdigit()):
    print(f"{P11str3}:- Valid")
else:
    print(f"{P11str3}:- Invalid")
```

User123:- Valid
 123User:- Invalid
 Us@12:- Invalid

12) Palindrome Checker (ignore the case)

- Madam → Palindrome
- Level → Palindrome
- Python → Not Palindrome

In [108...

```
P12str = "Madam"
P12str2 = "Level"
P12str3 = "Python"
P12str = P12str.lower()
P12str2 = P12str2.lower()
P12str3 = P12str3.lower()

ispalindrome = 0
for x in P12str:
    if (P12str == P12str[::-1]):
        ispalindrome = 1

if(ispalindrome == 1):
    print(f"{P12str} -> palindrome")
else:
    print(f"{P12str} -> Not palindrome")

ispalindrome = 0
for x in P12str2:
    if (P12str2 == P12str2[::-1]):
        ispalindrome = 1

if(ispalindrome == 1):
    print(f"{P12str2} -> palindrome")
else:
    print(f"{P12str2} -> Not palindrome")

ispalindrome = 0
for x in P12str3:
    if (P12str3 == P12str3[::-1]):
        ispalindrome = 1

if(ispalindrome == 1):
    print(f"{P12str3} -> palindrome")
else:
    print(f"{P12str3} -> Not palindrome")
```

madam -> palindrome
 level -> palindrome
 python -> Not palindrome

13) Longest Word Finder

- Python programming is interesting → programming
- I love coding → coding
- Data structures and algorithms → structures

```
In [1]: P13str = "Python programming is interesting"

words = P13str.split()
longest = max(words, key=len)

print("Longest word:", longest)
```

Longest word: programming

14) Case Conversion

- python programming
 - Upper: PYTHON PROGRAMMING
 - Lower: python programming
 - Title: Python Programming
 - Capitalize: Python programming

```
In [4]: P14str = "python programming"
print("Upper:",P14str.upper())
print("Lower:",P14str.lower())
print("Title:",P14str.title())
print("Capitalize:",P14str.capitalize())
```

Upper: PYTHON PROGRAMMING
 Lower: python programming
 Title: Python Programming
 Capitalize: Python programming

15) Custom Split

- apple,banana,grape
- 10|20|30
- A-B-C-D

```
In [8]: l1 ,l2 ,l3 = input("Enter A Values comma seprated").split()
print(l1,l2,l3 , sep=",")
n1 , n2 , n3 = input("Enter A Values comma seprated").split()
n1 = int(n1)
n2 = int(n2)
n3 = int(n3)
print(n1,n2,n3 , sep="|")
l4 ,l5 ,l6, l7 = input("Enter Four Letter: ").split()
print(l4 ,l5 ,l6, l7 , sep="-")
```

apple,banana,grape
 10|20|30
 a-b-c-d